University of Kansas

Group 18
CampusGear

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Software Architecture Document

1. Project Synopsis

CampusGear is a web application that allows KU students to reserve and borrow equipment while staff manage inventory, checkout, returns, and maintenance.

2. System Architecture Overview

2.1 High-Level Architecture Description

CampusGear will use a three-layer software architecture consisting of the Presentation Layer, Business Logic Layer, and Data Access Layer. This approach separates the website's user interface, application rules, and database operations into different components. The purpose of this design is to make the system easier to organize, develop, test, and maintain throughout the project.

Presentation Layer: This layer handles everything students and staff see and interact with on the website. It contains three main components: the Login Interface, Student Interface, and Staff Interface. Students can browse equipment, make reservations, and view their borrowing information. Staff can manage equipment, check items in and out, and monitor maintenance requests. The interface will be developed using HTML, CSS, and JavaScript.

Business Logic Layer: This layer manages the application's main functions and rules. It contains six components: User and Access Management, Equipment Management, Reservation Management, Loan Management, Maintenance and Damage Management, and Messaging and Notifications. These components check user permissions, manage equipment availability, prevent conflicting reservations, and process equipment checkouts and returns. Python with the Flask framework will be used to handle requests and coordinate these operations.

Data Access Layer: This layer manages communication with the SQLite database. It contains a Data Access Component responsible for saving, retrieving, and updating information such as user accounts, equipment records, reservations, loans, and maintenance reports. SQL queries will be used to interact with the database.

When a student or staff member performs an action, the request begins in the Presentation Layer and passes to the Business Logic Layer. The Business Logic Layer checks whether the action is allowed and communicates with the Data Access Layer when information needs to be retrieved or saved. The results are then returned through the Business Logic Layer to the Presentation Layer, where the website displays a confirmation, requested information, or an error message.

Using this architecture allows the team to develop different parts of CampusGear while keeping the components organized and connected.

2.2 Diagram 1

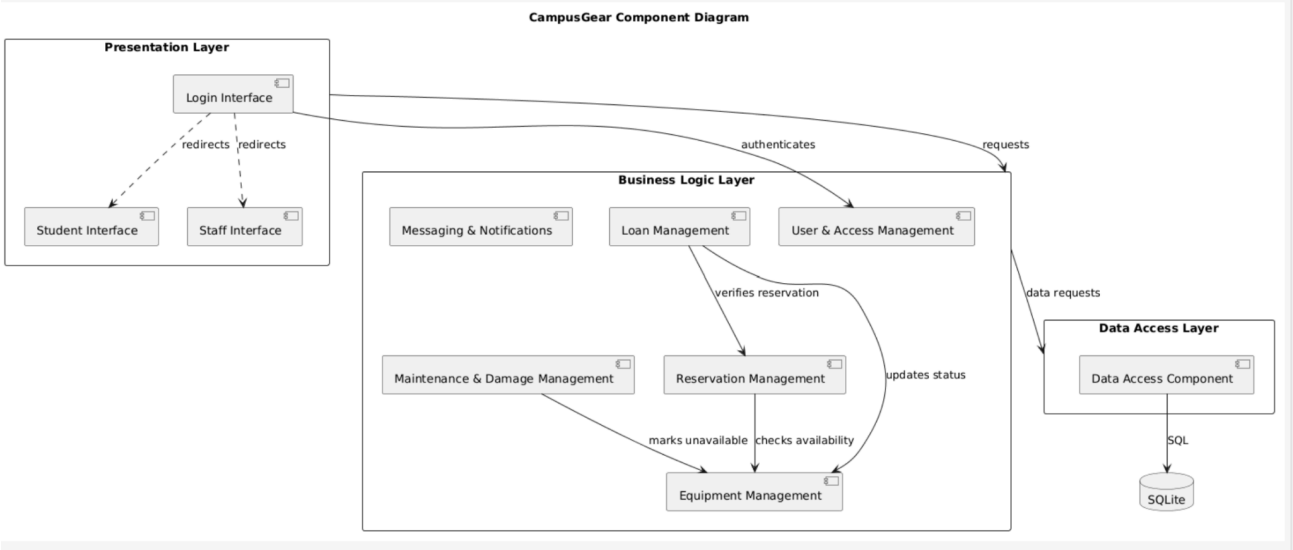


Diagram 1: CampusGear Component Diagram

The Login Interface authenticates users through User & Access Management and redirects them to the Student or Staff Interface. Both interfaces send requests to the Business Logic Layer, which sends data requests to the Data Access Layer, the only layer that queries the SQLite database.

2.3 Diagram 2

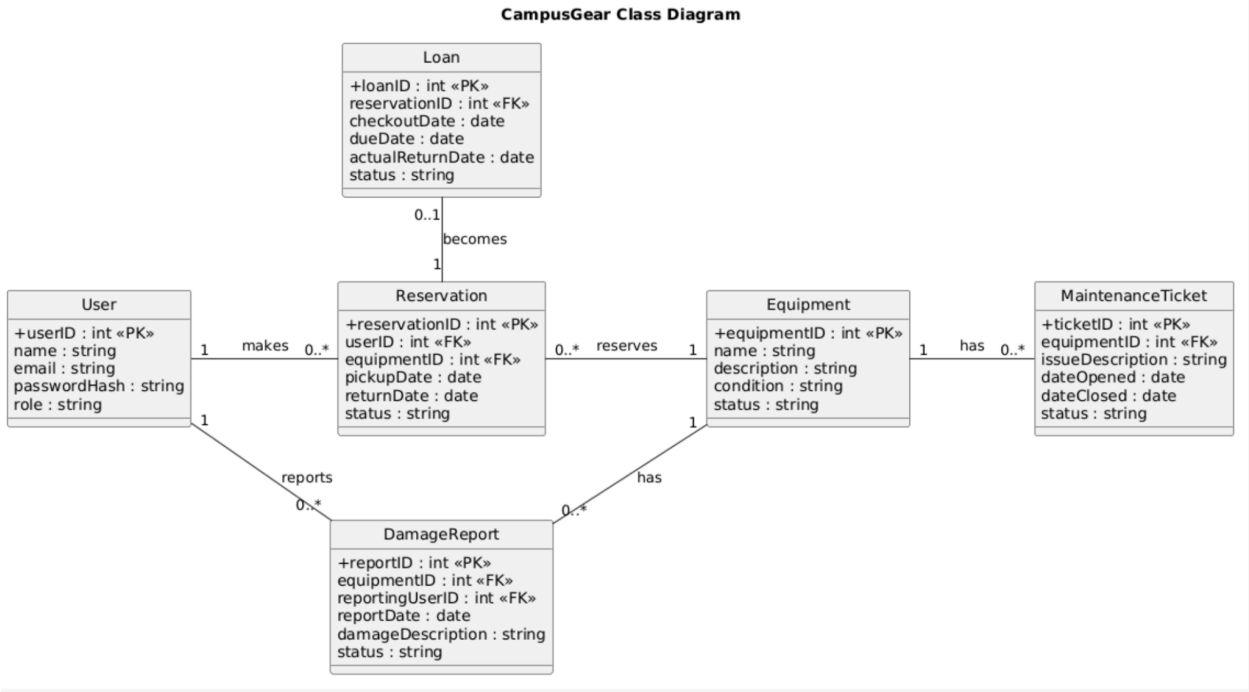


Diagram 2: CampusGear Class Diagram

This diagram shows the six main records stored in SQLite and how they connect through ID fields. A User makes many Reservations, and each Reservation is for one piece of Equipment. A Reservation becomes a

Loan only when staff check the item out at pickup, so it has zero or one Loan. A Loan is tied to its borrower and item through its Reservation. Each Equipment item can have many Maintenance Tickets and Damage Reports, and each Damage Report is submitted by one User.

2.4 Diagram 3

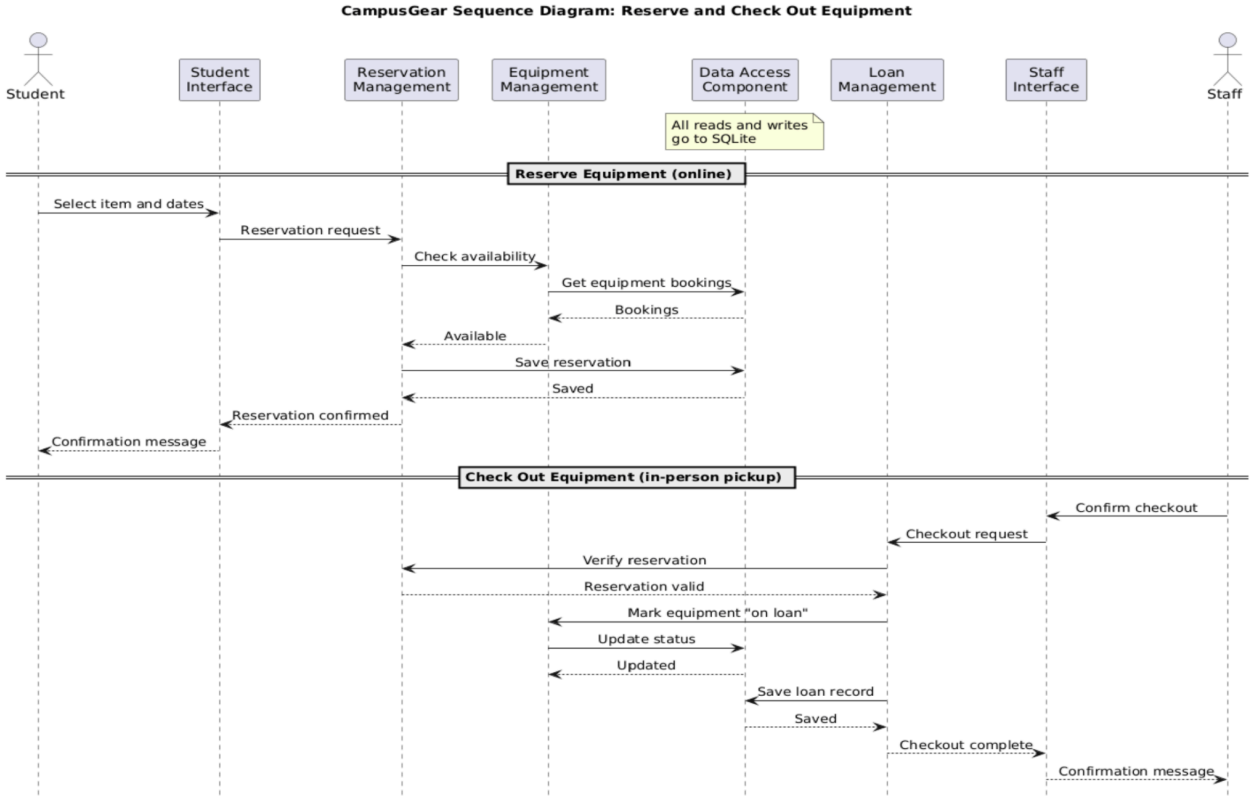


Diagram 3: CampusGear Sequence Diagram

This diagram shows the request flow for reserving and borrowing equipment. A student submits a reservation, and Reservation Management checks availability with Equipment Management before the Data Access Component saves it to SQLite and confirms the save. At pickup, staff confirm the checkout, and Loan Management verifies the reservation, has Equipment Management mark the item as on loan, and saves the loan record through the Data Access Component.

2.5 Diagram 4

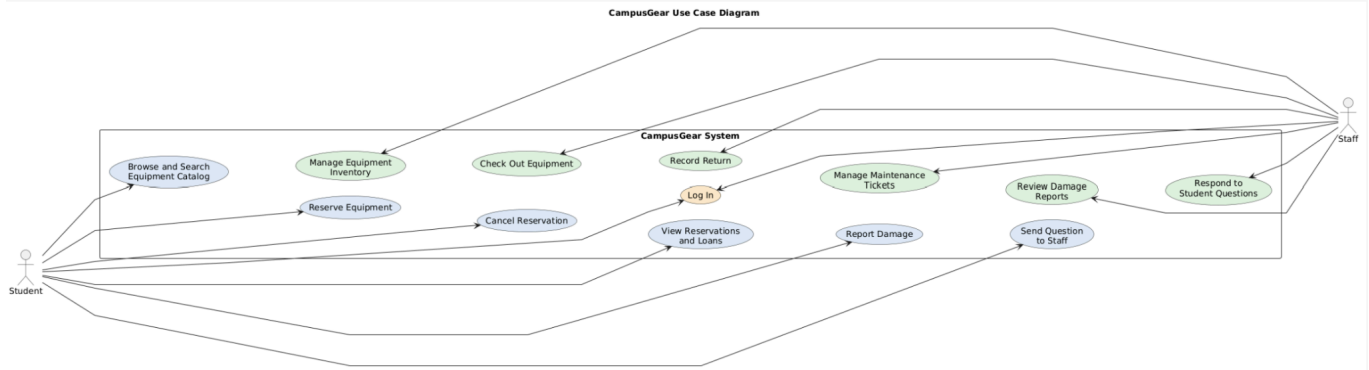


Diagram 4: CampusGear Use Case Diagram

This diagram shows the two types of users and the actions each can perform. Students and staff both log in. Students browse the equipment catalog, reserve and cancel equipment, view their reservations and loans, report damage, and send questions to staff. Staff manage inventory, check equipment out and in, manage maintenance tickets, review damage reports, and respond to student questions.

3. Architectural Components

3.1 Presentation Layer

3.1.1 Login Interface Component

Provides the login page where students and staff access CampusGear. Also, this component sends login information to the User and Access Management Component for authentication. After successful login, it directs the user to the Student or Staff Interface.

3.1.2 Student Interface Component

Provides students with access to equipment browsing, searching, reservations, borrowing history, due dates, reservation cancellations, damage reporting, and student support features. This component sends equipment queries to the Equipment Management Component, reservation requests to the Reservation Management Component, and borrowing-related requests to the Loan Management Component. Also, this component communicates with the Maintenance and Damage Management Component and the Message and Notifications Component when students report damage or need to contact the staff.

3.1.3 Staff Interface Component

Provides staff with interfaces for managing inventory, processing equipment checkouts and returns, handling maintenance, and responding to student questions. Also, this component communicates requests to the Equipment Management, Loan Management, Maintenance and Damage Management, and Messaging and Notifications components, while displaying the results returned by those components.

3.2 Business Logic Layer

3.2.1 User and Access Management Component

Authenticates users and manages role-based permissions for students and staff. This component receives login requests from the Login Interface Component and works with the Data Access Component to retrieve or update user account records.

3.2.2 Equipment Management Component

Manages equipment inventory, including registering items, updating equipment details, search the catalog, and tracking availability. Also, this component communicates with the Reservation Management and Loan Management Components to ensure correct equipment status and with the Maintenance and Damage Management Component to recognize when equipment is under repair.

3.2.3 Reservation Management Component

Handles equipment reservations, cancellations, availability checks, and waitlists. This component also communicates with the Equipment Management Component to check availability, the Loan Management Component when reserved equipment is checked out, and the Data Access Component to store and retrieve reservation records.

3.2.4 Loan Management Component (checkouts, returns, due dates)

Manages equipment checkouts, returns, due dates, and loan extensions. Also, this component communicates with the Reservation Management Component to verify reservations and the Equipment Management Component to update equipment availability. The Data Access Component helps to store and update the loan records.

3.2.5 Maintenance and Damage Management Component

Records damage reports, creates maintenance tickets, and tracks equipment repairs. This component communicates with the Equipment Management Component to mark damaged equipment unavailable and

update its status when repairs are finished. The Data Access Component helps to store and retrieve maintenance records.

3.2.6 Messaging and Notifications Component

Allows students to submit questions and staff to record responses. It can also provide notifications about relevant reservation updates and upcoming due dates. This component communicates with the Student and Staff Interface components and uses the Data Access Component to store and find messages and needed records.

3.3 Data Access Layer

3.3.1 Data Access Component

Handles database operations for users, equipment, reservations, loans, maintenance tickets, and messages. This component receives data requests from the business logic components, executes SQL queries against the SQLite database, and returns retrieved records or results.

4. Key Data Structures

CampusGear stores its main records in SQLite. Each record has a unique ID, and related records use these IDs to connect users, equipment, reservations, and loans.

User: Stores account information for a student or staff member. Important fields include user ID, name, email address, password hash, and role. The role determines whether the user can access student or staff features.

Equipment: Represents one physical school rental item or complete kit, such as a Raspberry Pi or soldering kit. Fields include equipment ID, name, description, condition, and status, such as available, on loan, or under maintenance. Separate items have separate IDs, even when they are the same type of equipment.

Reservation: Records an online booking that holds an item for a student to pick up in person. The reservation itself does not start the loan. Fields include reservation ID, user ID, equipment ID, pickup date, return date, and reservation status. The user and equipment IDs connect the reservation to the borrower and the selected item. Reservation dates help the system check for overlapping bookings.

Loan: Tracks the borrowing period, which starts when staff check out the item at physical pickup and ends when the student returns it in person. Fields include loan ID, reservation ID, checkout date, due date, actual return date, and loan status. The reservation links the loan to its user and equipment. The actual return date remains empty until the item is returned.

Maintenance Ticket: Tracks an equipment issue that staff need to address. Fields include ticket ID, equipment ID, issue description, date opened, date closed, and ticket status. The equipment ID identifies the affected item, and the date closed remains empty until the work is finished.

Damage Report: Records damage reported for an item. Fields include report ID, equipment ID, reporting user ID, report date, damage description, and report status. These records help staff review the damage and determine whether the item needs maintenance.

5. Interface Description

5.1 User Input

Users enter information in CampusGear to log in, reserve equipment, manage equipment details, confirm checkouts and returns, report damage, and manage maintenance tickets. To log in, they enter their email address and password. The backend checks these details against user accounts stored in SQLite, where passwords are stored as hashes.

Students reserve school rental equipment online by choosing an item from the catalog and entering pickup and return dates before visiting the pickup location. Staff can add or update an item’s name, description,

and availability. At in-person pickup, staff select the student's reservation and confirm checkout to start the loan. When the student returns the item in person, staff record the return. To report damage, users select the affected item and enter a description of the damage. Staff review these reports and update their status. To create a maintenance ticket, staff select an item and describe the issue. They can update the ticket's status and close it when the work is finished. Equipment under maintenance is marked as unavailable for borrowing.

The frontend checks that required fields are filled in and dates use a valid format. The Flask backend checks the submitted information as well. It makes sure the return date is after the pickup date and that the equipment is available during the requested dates. It also checks that anyone performing a staff action has an authorized staff account. If information is missing or invalid, the system shows a clear error message and doesn't save the submission. Valid changes are saved in SQLite, and a confirmation message lets the user know the action was successful.

5.2 User Interface

Users access CampusGear through a web browser. After logging in, they see navigation options for their role as a student or staff member. Students can browse a catalog of school rental equipment, such as soldering kits and Raspberry Pis, showing each item's name, description, and availability. For available equipment, they can open an online reservation form and choose dates for in-person pickup and return.

Students can use the "My Reservations and Loans" page to view their online reservations and active loans, including the equipment, pickup and return dates, and current status. This page distinguishes equipment reserved for pickup from equipment already checked out. A simple damage-report form lets users select an item and describe the damage. Staff use a dashboard to add or update equipment and record checkouts and returns. The dashboard also lists damage reports and maintenance tickets with their equipment details, issue descriptions, and current status. Staff can review reports, create or update maintenance tickets, and mark work as finished. The equipment catalog shows when an item is under maintenance and unavailable for borrowing.

Each page uses clear labels for fields, buttons, and navigation links. Confirmation messages let users know when an action succeeds, and error messages explain what they need to fix.

6. Architectural Constraints and Assumptions

Frontend — HTML, CSS, JavaScript

- HTML: Structure of pages, forms, tables, and equipment listings.
- CSS: Layout, styling, and responsive design.
- JavaScript: Interactive elements, form validation, and communication with the backend.

Backend — Python with Flask

- Handles reservation rules, equipment management, checkouts, returns, and role permissions.
- Flask is a lightweight web framework that is relatively easy to learn and set up.
- Provides endpoints for the frontend to retrieve or update data.

Database — SQLite with SQL

- Stores users, equipment, reservations, loans, maintenance tickets, and damage reports.
- SQL is the language used to query and update the database.
- SQLite is lightweight and doesn't require a separate database server.